Part A Chapter 9

Reactive Behaviour[1]

1 Introduction

An animal or organism (which also can be human) has simple reactive behaviour if it immediately responds to stimuli from its environment. Such a pattern of behaviour is called *stimulus-response behaviour*. Animals that show this behaviour make their decisions based on information they receive as input, and simple situation-action associations. The stimuli they get as input can either consist of observed changes in the external world (by perception, sensing) or received communications from other animals. The behaviour of the animal in response affects the environment. Another variant of reactive behaviour is *delayed response behaviour*, where also stimuli in the past may affect the animal's decisions about its actions. In this chapter both types of behaviour will be analysed and modelled from an external behavioural view and from an internal cognitive view.

1.1 Domain description and focus

The explorations in behaviour in this and the next chapter are illustrated by an example, taken from the discipline that studies animal behaviour; e.g., (Vauclair, 1996). Many experiments with animal behaviour with respect to food have been performed and reported during the last century. Animals, for example dogs, sometimes show a *delayed response*: they look for food in places where they have seen food before, even if they do not see it at that moment. This suggests that these animals might have some internal representation or memory of stimuli received earlier. More systematic experiments on this delayed response issue, for example those reported in (Hunter, 1912; Tinklepaugh, 1932), support this suggestion.

An Experimental Setting

The type of experiment reported in (Tinklepaugh, 1932) is set up as follows (see Figure 1). Separated by a transparent screen (a window, at position p0), at each of two positions p1 and p2 a cup (upside down) and/or a piece of food can be placed. At some moment (with variable

[1] Based on: Jonker, C.M., and Treur, J., Agent-Based Simulation of Animal Behaviour. *Journal of Applied Intelligence,* vol. 15, 2001, pp. 83-115.

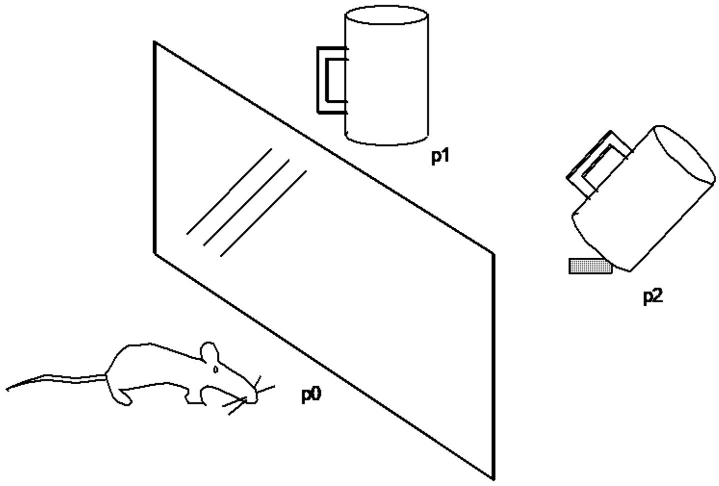

Figure 1 Situation 3 of the experiment

delay) the screen is raised, and the animal is free to go to any position. Within this experiment the animal is able to observe the positions of food, cups, screen, and itself. The following four possible situations are considered:

Situation 1 At both positions p1 and p2 an empty cup is placed.

Situation 2 At position p1 an empty cup is placed, and at position p2 a piece of food, which is (and remains) visible for the animal.

Situation 3 At position p1 an empty cup is placed and at position p2 a piece of food is placed, after which a cup is placed at the same position, covering the food. After the food disappears under the cup the animal cannot sense it anymore.

Situation 4 At position p1 an empty cup is placed and at position p2 a cup and a piece of food are placed, in such a manner that the animal did not see the food.

After the screen has been taken away, in situation 1 the animal, not having observed any food, will not show a preference for either position p1 or p2; it may even go elsewhere or stay where it is. In situation 2 the animal, observing food at position p2 and that no screen is present, will go to position p2, which can be explained as pure stimulus-response behaviour. In situation 3 the immediate stimuli are the same as in situation 1: no food is observed, and it is observed that no screen is present. Animals that react in a strictly stimulus-response manner will respond to this situation as in situation 1. Animals that show delayed response behaviour will go to p2, where food can be found, as has been observed in the past. In situation 4 animals will behave as in situation 1; however, here they can find food and eat it, and therefore be reinforced in successful behaviour.

In the literature, many reports can be found of observed delayed response behaviour in experiments of the type described above: (Vauclair, 1996), p. 4-5. The animal species used in these experiments vary from rats and dogs to macaques, chimpanzees and human infants. Therefore, it is assumed that animals of the type studied maintain internal (mental) representations (a world state model) of the world state on the basis of their sensor input, and that they make use of this world state model (in addition to the actual sensory input that is used) to determine their behaviour.

1.2 Characteristic patterns to be explored

Four possible patterns of behaviour A, B, C, and D for the experiment can be described. Two of them (A. and B.) are addressed in this chapter, the other two in the next chapter. The following dynamic relationships of their behaviour express possible hypotheses that can be made about the behaviour of animals in the experiments:

A. Stimulus-response behaviour

The animal behaves the same for the situations 1, 3 and 4 described above: doing nothing, as if no food is present. Only in situation 2 it goes to the position of the food.

B. Delayed response behaviour

The animal behaves the same in the situations 2 and 3: it goes to the position of the food. In situations 1 and 4 it does nothing.

C. Motivation-based behaviour

The animal's behaviour in the situations 1, 2, 3 and 4 depends on whether the animal has a motivation or desire to do so. For example, the animal may start acting in a pro-active manner (without any specific stimulus) in situation 1.

D. Adaptive behaviour

The animal is able to adapt its behaviour on the basis of experiences. For example, if the cups have different colours and a certain colour is used more often for cups under which food is hidden, then after some learning time the animal looks with priority for food at positions with a cup of that colour. In a similar manner an animal may learn from experience that at a certain position more often food can be found.

In summary, within subsequent chapters different patterns of behaviour are addressed. Based on the experimental setting discussed above, patterns of reactive behaviour are addressed in this chapter; in particular, stimulus-response behaviour and delayed response behaviour. Motivation-based behaviour is the subject of Part B, Chapter 3, including the attribution of motivational concepts to (other) animals or humans. Adaptive behaviour is addressed in Part B, Chapter 4.

Stimulus-response behaviour patterns

The four simple example traces in Table 1 for situation 1 (the situation without cups) illustrate the stimulus-response pattern of behaviour within the experimental setting. Here the time points indicated are time points that show a different state, so no uniform time scale is assumed. The observed animal receives observation input on the availability of food (indicated by food), and of the limitation of its moving around due to the presence or absence of a screen in the experimental setting (indicated by screen).

time *trace*	*time point 0*	*time point 1*	*time point 2*	*time point 3*	*time point 4*	*time point 5*
trace 1	food at p2 screen	no food at p2 screen	food at p2 screen	food at p2 no screen	food at p2 no screen goes to p2	no food at p2 no screen
trace 2	no food at p2 screen	no food at p2 screen	no food at p2 screen	food at p2 screen	food at p2 no screen	food at p2 no screen goes to p2
trace 3	no food at p2 screen	no food at p2 no screen	no food at p2 no screen	food at p2 no screen	food at p2 no screen goes to p2	no food at p2 no screen
trace 4	food at p2 screen	no food at p2 no screen	food at p2 screen	food at p2 screen	food at p2 no screen	food at p2 no screen goes to p2

Table 1 Example set of observed stimulus-response traces

Assume that the traces depicted in Table 1 are observed. In this table, for example, the fact that at a certain point in time it is observed that there is food at p2 is simply denoted by food at p2, and goes to p2 denotes that the animal performs the action to go to p2.

Delayed response behaviour patterns

In delayed response behaviour, previous observations may have led to internal maintenance of some form of memory of the world state, a model or representation of the (current) world state (for short, *world state model*). This form of memory can be used at any point in time as an additional source (in addition to the direct observations) to determine behaviour. In that case the same input pattern of stimuli can lead to different behaviour, since the world state models based on observations in the past are different. This type of behaviour, which just like stimulus-response behaviour occurs quite often in nature, is a bit more complex than stimulus-response behaviour. This leads to the question what kind of complexity in the environment is coped with by this kind of behaviour that is not coped with by stimulus-response behaviour. An answer on this question can be found in a type of environment with aspects which are important for the animal (e.g., food or predators), and which cannot be completely observed all the time; e.g., food is sometimes covered by other objects.

time *trace*	*time point 0*	*time point 1*	*time point 2*	*time point 3*	*time point 4*	*time point 5*
trace 1	food at p2 screen	cup at p2 screen	cup at p2 screen	cup at p2 no screen	cup at p2 no screen goes to p2	cup at p2 no screen
trace 2	no food at p2 screen	no food at p2 screen	cup at p2 screen	cup at p2 screen	cup at p2 no screen	cup at p2 no screen
trace 3	no food at p2 screen	no food at p2 no screen	no food at p2 no screen	no food at p2 no screen	food at p2 no screen	food at p2 no screen goes to p2
trace 4	food at p2 screen	no food at p2 screen	food at p2 screen	cup at p2 screen	cup at p2 no screen	cup at p2 no screen goes to p2

Table 2 Example set of observed delayed response traces

In Table 2 some example traces of delayed response behaviour are depicted. Here, only food is depicted when it is observable. The following questions are addressed below (see also Table 1 in Part B, Chapter 1):

- Which observations (input) and actions (output) can be used to describe this externally observed behaviour?
- What is the external behavioural pattern behind this observed behaviour; which external dynamic relationships between input and output states can be expressed that characterize the pattern behind these traces?
- Which assumed internal cognitive state properties generate this externally observed behaviour?
- What is the pattern of dynamics of these internal cognitive state properties; how can these internal dynamics be characterized by dynamic relationships?

2 Analysis of the Main Aspects and Their Relations

In this section the main concepts and their relations needed for the model are identified.

2.1 Identification of relevant concepts

To describe reactive behaviour, as input *observations* are used, in particular of the presence of *food* at a certain position, and of the *screen*. Moreover, at the output, *actions* are used, in particular to *go to* a certain position. Internal concepts used are *sensory representations* and *beliefs*. Sensory representations are internal states (for example, realised as activations of neurons) correlating to observations, occurring for a short time interval immediately after the moment of observation. Beliefs store acquired information over longer time periods. Both concepts can be used for information about the presence of food or the screen. In summary, the following concepts are relevant:

External behavioural view

- input: observation (of presence of food or screen)
- output: action (goto p2)

Internal cognitive view

- sensory representation (of presence of food or screen)
- belief (of presence of food or screen)

2.2 Global structure of the model

The model is specified at two levels: the external behaviour level, and the internal, cognitive level. A global overall picture of the model can be found in Figure 2.

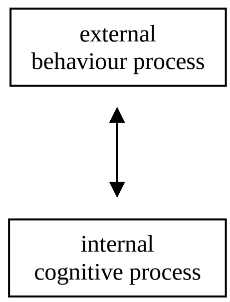

Figure 2 Global structure of the model

Interaction between the levels can take place by transferring input from the behavioural level to the cognitive level, determining output by the cognitive process, and transferring this output back to the behavioural level. Alternatively, the model at the behavioural level can also be used to determine the output states from the input states (i.e., abstracting from the underlying cognitive level).

2.3 Relationships between the identified concepts

The relationships between the identified concepts are as follows; see also Figures 3, 4 and 5. Notice that the box indicates the borderline between the internal and externally visible parts.

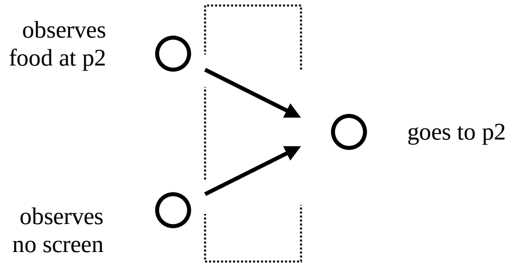

Figure 3 External behavioural view on stimulus-response behaviour

Relationships at the external behavioural level for stimulus-response behaviour

- an observation that there is food at p2 affects that it will go to p2.
- an observation that no screen is present affects that it will go to p2.

Relationships at the internal cognitive level for stimulus-response behaviour

- an observation that there is food at p2 affects sensory representation for food
- an observation that no screen is present affects sensory representation for no screen
- a sensory representation for food affects that the animal will go to p2.
- a sensory representation for no screen affects that the animal will go to p2.

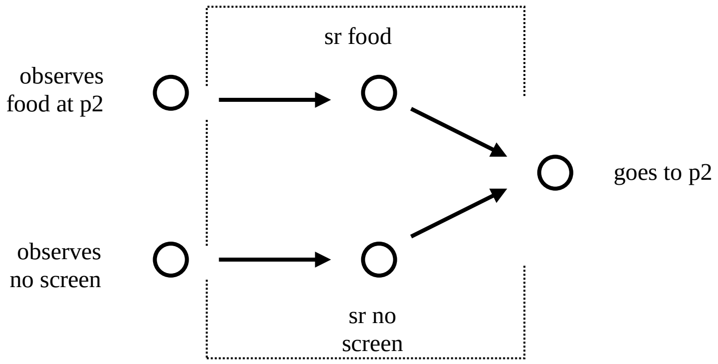

Figure 4 Internal cognitive view on stimulus-response behaviour

Relationships at the external behavioural level for delayed response behaviour

- an observation that no screen is present affects that it will go to p2.
- an observation in the past that there was food at p2 affects that it will go to p2.

Note that the latter type of relation between states over a longer time period cannot be visualised in the form of a picture with arrows indicating direct effects of states on each other.

Relationships at the internal cognitive level for delayed response behaviour

Since the external characterisations of delayed response behaviour refer to the animal's input in the past, it makes sense to assume that the animal makes use of some mechanism to maintain past observations (in the form of a certain kind of *world state model*) by means of persisting internal properties, i.e., some form of memory. These persisting properties are sometimes called *beliefs* on the world state. For the example case, assume that an internal belief that food is present at position p2 is available that defines part of the animal's world state model, so that the following dynamic relationships hold:

- an observation that food is present at position p2 affects the belief that food is present at position p2
- the belief that food is present at position p2 affects the belief that food is present at position p2 at later time points
- an observation that no screen is present affects that the animal will go to position p2
- the belief that food is present at position p2 affects that the animal will go to position p2

In graphical form these relationships can be depicted as shown in Figure 5.

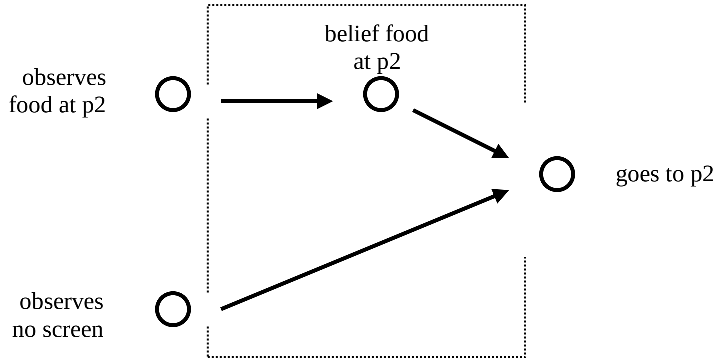

Figure 5 Internal cognitive view on delayed response behaviour

3 The Detailed Model

In this section the detailed model is introduced. First it is shown in which way the relevant concepts are formalised, and next the dynamic relations are formalised in a detailed manner.

3.1 Formalising the concepts used

To obtain a detailed model, first the concepts used are formalised. In this case for each of the concepts a logical atom is introduced. These atoms are based on predicates for observation, performance, sensory representation and belief.

External behavioural level		
observation	(of presence of food or screen)	observed(at_position(food, p2)) observed(screen)
action	(goto p2)	performed(goto(p2))
Internal cognitive level		
sensory representation	(of presence of food or screen)	sr(at_position(food, p2)) sr(screen)
belief	(of presence of food or screen)	belief(at_position(food, p2)) belief(screen)

Table 3 Formalisation of concepts used

3.2 Formalising the relationships

The dynamics of these concepts involve temporal relationships, which are analysed in more detail below. They describe the basic parts of the process. To formalise the temporal relationships whenever possible, an executable format is used.

3.2.1 Temporal relationships in executable format

Temporal relationships in executable format are defined as follows.

a) A dynamic relationship is called *a step relationship* if and only if it has the form:

if	X	holds,
then	Y	will hold

Here X and Y are (combinations of) state properties: properties of states such as ‘observes a screen’, ‘performs the action of going to p2’, or ‘sensory representation of the food at p2’.

b) A dynamic relationship is called *a (conditional) persistence relationship* if and only if it has the form:

if	X	holds
then	X	will hold at all later time points, as long as not Y holds

Here, as in a), X and Y are (combinations of) state properties.

c) A dynamic relationship is in *executable format* if it is either specified as a step relationship or as a persistence relationship.

The ‘will hold’ in the consequent in a) and b) expresses that this will be the case some time duration after the time point which is considered for the antecedent. This time duration can be left unspecified, or can be made explicit in the form of some real or natural number. In the notation for executable properties often the symbol $\twoheadrightarrow$ for ‘leads to’ (or LEADSTO) will be used. In formal logical notation an example step relationship would look like:

$X1 \wedge X2 \twoheadrightarrow Y$ (X1 and X2 together lead to Y)

A conditional step relationship would look like:

$X \wedge \neg Y \twoheadrightarrow X$ (X persists unless Y occurs)

Note that persistence in b) can also be defined for one step; if such a persistence step is repeated a number of times, the effect is the same as that of a persistence relationship defined as in b) above. The notion ‘executable format’ defined in this manner is a rather simple format. Dynamic relationships in the simple executable format defined above have five advantages:

(a) they can be depicted in a transparent manner using a simple graphical format
(b) they allow for direct simulation or implementation in computational context.
(c) they allow for realisation in natural (physical, chemical, fysiological, neurological) context by causal relations.
(d) they allow for explanations of a particular action of an animal based on elementary computation steps leading to the action
(e) they allow for explanation of the external behavioural process description based on underlying elementary (e.g., causal) mechanisms

Concerning (a), a graphical format that can be used to depict a step relationship is shown in Figure 6. Note that this does not provide a complete specification as it is not represented whether X1 and X2 together lead to Y or whether they each lead to Y. This is similar to the case of modelling by numerical variables, where such a picture does not specify by which calculation precisely the value of Y can be determined from the values of X1 and X2.

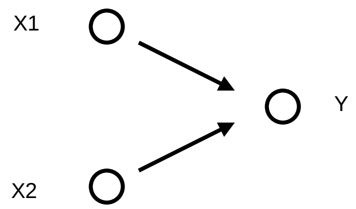

Figure 6 Graphical representations of executable temporal relationships

Concerning (b) above, within a computational context a step relationship can be implemented as a simulation or transition step computing the next state out of the current state. The persistence relationship can be implemented by storage in a memory. Concerning (c), within a physical context, a step relationship can take the form of a causal relation (X causes Y), whereas a persistence relationship is a basic property of nature (in absence of any specific causal effects).

To obtain (d), any particular action an animal performs can be explained by tracing back the steps using the executable dynamic relationships of the internal cognitive perspective. Moreover, for (e), the general description of the external dynamics is entailed by the set of internal executable relationships, and therefore can be explained from them. As these executable relationships can be related to either elementary computation steps or basic causal relations in nature, explanations can be generated for either a computational or natural (physical) context.

3.2.2 Relationships at the external behavioural level for stimulus-response behaviour

A relationship from the external perspective can be expressed informally as follows:

> Always if the animal observes that there is food at p2, and no screen is present,
> it will go to p2.

To express such dynamic relationships in a more standardized manner, a structured semi-formal temporal language will be used. This language will also turn out useful for more complex patterns of behaviour, as will be seen later in this and subsequent chapters. Moreover, such a structured form makes it easier to formalise the relationship, for example, in the form of a temporal LEADSTO relation. For example, for the fact that a given trace shows stimulus-response behaviour, the above informally stated relationship can get a structured textual form and formal form of an executable temporal relationship expressed as follows.

ESR1

at any point in time,
if the animal observes that food is present at position p2
and it observes that no screen is present,
then it will go to position p2

In formal language:

observed(at_position(food, p2)) $\wedge$ observed(no_screen) $\twoheadrightarrow$ performed(goto(p2))

The naming convention of temporal relationships like these is as follows: the first letter refers to whether it is an external or internal relationship, the rest is a label characterizing the type of behaviour. Notice that this relationship indeed relates an input state (in which the conditions expressed in the antecedent hold) to an output state (in which the consequent holds).

Notice that phrases such as 'observes no screen' means that the absence of the screen is observed, so it does not mean that no observation takes place. The (epistemic) difference between 'not observing whether it rains' and 'observing that it does not rain' is similar to the difference between 'not knowing whether it rains' and 'knowing that it does not rain'.

Explanation of stimulus-response behaviour from an external behavioural perspective

Given the external behavioural description specified above, the following question concerning explanation can be put forward.

- How can observed behaviour be explained using an external behavioural description ?

To make it more specific, assume it is observed that the animal goes to p2. The type of explanation we consider has the following form:

Why does the animal go to p2 ?

The animal goes to p2 because it observed that food is present at p2, and that the screen is absent.

The dynamic relationship **ESR1** as defined leaves completely open the response time. An animal going to p2 next day, or next year will also count as an instance of stimulus-response behaviour. A question is what would be a reasonable maximal response time for a response to count as a response to a given stimulus. Probably this maximal response time depends on the case at hand. For an animal in our experimental setting, a maximal response time of 1 to 4 seconds might be reasonable. But in other cases this may be too long or too short (imagine the response time of a snail). Within the dynamic relationship above, it is not difficult to include a parameter for the maximal response time taken into account. Thus more refined variants of the dynamic relationship can be made, for example, taking into account a maximal response time r; in structured semiformal form this obtains the following parameterised (by r) relationship:

ESR2(r)

at any point in time,
if the animal observes that food is present at position p2
and it observes that no screen is present,
then within at most r seconds it goes to position p2

3.2.3 Relationships at the internal cognitive level for stimulus-response behaviour

Although an adequate external description of a pattern of stimulus-response behaviour is possible, this does not exclude to consider the possibility of a description from the internal perspective. For this example, internal cognitive concepts in the form of sensory representations for food present and screen absent can be assumed with the following dynamic relationships:

ISR1

at any point in time,
if the animal observes that food is present at position p2
then internal sensory representation for food at p2 will hold

ISR2

at any point in time,
if the animal observes that no screen is present,
then internal sensory representation for no screen present will hold

ISR3

at any point in time,
if internal sensory representation for food present at p2 holds
and internal sensory representation for no screen present holds,
then the animal will go to position p2

In formal language:

observed(at_position(food, p2)) ↠ sr(at_position(food, p2))

observed(no_screen) ↠ sr(no_screen)

sr(at_position(food, p2)) ∧ sr(at_position(food, p2)) ↠ performed(goto(p2))

Notice that for cases that the observations are of a very short duration, the phrases ‘sensory representations for food present will hold’ and ‘sensory representations for screen absent will hold’ could be true at different points in time, and thus never be satisfied simultanuously. In such a case the relationships would not lead to the action. For this description of internal dynamics to work, either the timing of these conditions need to be defined by an exact time delay after the (simultanuous) observations, or they need to persist sufficiently long, so that both sensory representations occur (at least) at a common point in time.

3.2.4 Relationships at the external behavioural level for delayed response behaviour

In an informal manner, a dynamic input-output relationship of delayed response behaviour can be expressed as follows.

> Every time that the animal observes that no screen is present,
> and it observed in the past that there was food at p2,
> it will go to p2.

Using more structured semi-formal temporal language, for the example experimental setting, the input-output relationship for delayed response behaviour can be characterised as:

EDR1

at any point in time,
if it observes that no screen is present
and at some earlier point in time the animal observed that food was present at position p2,
then the animal will go to position p2

As in this case the temporal relationship is a bit more complex than the previous types of dynamic relationships, no graphical representation is given for this relationship; the same holds for relationships **EDR2** and **EDR3** below. The graphical representation can be used for relationships in executable format, as discussed in Section 3.2.1, but the relationships mentioned are not in executable format.

Explanation of delayed response behaviour from an external behavioural perspective

Also for this case the question about explanation can be posed.

- What does an explanation look like if this description is available ?

Consider the following example explanation.

> Why does the animal go to p2 ?
> The animal goes to p2 because it just observed that no screen is present, and in the past it observed that food was present at p2.

This explanation refers back to the past. This past event cannot be seen as a direct cause of the current behaviour. The question then becomes the following.

- What is, besides the current observation of the absence of the screen, a more direct cause than the reference to observation of food in the past ?

This question cannot be answered easily from the external behavioural perspective. Below this question will be answered from the internal cognitive perspective.

There is at least one course of affairs in which dynamic relationship **EDR1** provides debatable results as a specification of the dynamic input-output relationship of delayed response behaviour. Suppose the animal observes food while the screen is there. After this another animal enters the scene and eats the food, which makes the food disappear. All this is observed by our animal. After this observation a cup is put at p2. Then the screen is taken away, and the animal goes to p2. This example trace, and some similar ones are depicted in Table 4.

time *trace*	*time point 0*	*time point 1*	*time point 2*	*time point 3*	*time point 4*	*time point 5*
trace 1	food at p2 screen	no food at p2 screen	no food at p2 screen	cup at p2 screen	cup at p2 no screen	cup at p2 no screen goes to p2
trace 2	food at p2 screen	no food at p2 screen	cup at p2 screen	no food at p2 screen	no food at p2 no screen	no food at p2 no screen goes to p2
trace 3	no food at p2 screen	food at p2 screen	no food at p2 screen	no food at p2 no screen	no food at p2 no screen	no food at p2 no screen goes to p2

Table 4 Example set of debatable traces

Although after these events the animal's most recent observation is that no food is present at p2, according to the relationship expressed above, still the animal will go to the position p2, because both conditions are fulfilled. The second and third trace in Table 4 are even more challenging. For example, in trace 3, no cup is involved at all. The animal observes that no food is present all the time after time point 2 (and it may even have eaten the food itself, or moved the food to a different position!), but still goes to p2. It may do so for whatever reason, but what is debatable is that the dynamic relationship expressed above forces the animal to go to p2 in this case. This happens because this formulation of delayed response does not take into account whether or not since the food was observed, the food has disappeared and this absence of food has been observed as well. It may be the case that such a pattern of behaviour occurs in nature: *rigid delayed response behaviour*.

The considerations above make it worth while yet to look for another, more flexible form of delayed response behaviour. The dynamic input-output relationship of such a pattern of behaviour can be specified as a temporal relationship, which excludes phenomena as discussed above. In an informal manner, such a more sophisticated dynamic relationship of, *updating delayed response behaviour*, can be expressed as follows.

> Every time that the animal observes that no screen is present, and it observed in the past that there was food at p2, and it did not observe after this time point that there was no food at p2, it will go to p2.

Using semi-formal temporal language, this dynamic input-output relationship characterizing the pattern of updating delayed response behaviour can be expressed as follows:

EDR2

at any point in time t1,
if the animal observes that no screen is present
and at some earlier point in time t the animal observed that food was present at position p2,
and at every point in time t' after t up to t1,
the animal did not observe that no food was present at p2,
then the animal will go to position p2

This relationship does not force the (debatable) behaviour depicted in Table 4. Other, more refined input-output relationships can be expressed as well. First, it is possible to limit the response time by a bound *r* after it was observed that no screen is present. Another refinement is to limit the past observation by some bound *b*. Using these parameters, a more refined relationship can be expressed as follows in a structured semi-formal manner:

EDR3(b, r)

at any point in time t1,
if the animal observes that no screen is present
and at some earlier point in time t ≥ t1–b the animal observed that food was present at position p2,
and at every point in time t' after t up to t1,
the animal did not observe that no food was present at p2,
then within at most r seconds it goes to position p2

3.2.5 Relationships at the internal cognitive level for delayed response behaviour

In Section 3.2.4 above, dynamic relationships were established that characterize the dynamic input-output relationship of delayed response behaviour from an external behavioural viewpoint; these relationships refer to the animal's input in the past, in relation to the animal's present output. In this section the internal cognitive viewpoint is taken. Assumptions are made about the existence of internal cognitive state properties and internal dynamics in such a way that these internal dynamics produce the externally characterized delayed response behaviour. To characterize these internal dynamics, dynamic relationships are identified that not only refer to input states and output states over time, but also to *internal cognitive states*.

Since the external characterisations of delayed response behaviour refer to the animal's input in the past, it makes sense to assume that the animal makes use of some mechanism to maintain past observations (in the form of a certain kind of *world state model*) by means of persisting internal properties, i.e., some form of memory. These persisting properties are sometimes called *beliefs* on the world state. For the example case, assume that an internal belief that food is present at position p2 is available that defines part of the animal's world state model, with the following dynamics:

IDR1

for all time points
if the animal observes that food is present at position p2,
then internal belief that food is present at position p2 will hold

IDR2

for all time points
if internal belief that food is present at position p2 holds,
then for every later time point internal belief that food is present at position p2 holds

IDR3

at any point in time,
if it observes that no screen is present,
and internal belief that food is present at position p2 holds,
then the animal will go to position p2

In formal language

observed(at_position(food, p2))	↠	belief(at_position(food, p2))
belief(at_position(food, p2))	↠	belief(at_position(food, p2))
belief(at_position(food, p2)) ∧ observed(no_screen)	↠	performed(goto(p2))

Informally stated, relationship **IDR1** expresses that any observation of p2 leads to the belief that food is present at position p2. The second relationship **IDR2** expresses that once this belief holds, it will persist forever. The third relationship **IDR3** defines that the animal will go to p2 if the belief that food is present at p2 holds and it observes that no screen is present. Variants of these relationships can be made that take into account a maximal delay time.

Explanation of delayed response behaviour from an internal cognitive perspective

An internal cognitive perspective takes into account internal cognitive state properties. In this way an animal's action can be described and explained by relating it to internal cognitive state properties preceeding the action. In Section 3.2.4 the situation was encountered that an explanation from an external behavioural perspective requires a reference to an observation of food in the past, and thus did not offer a direct cause of the behaviour. So:

- How would an explanation look like from the internal cognitive perspective ?

The following example explanation illustrates the issue.

Why did the animal go to p2 ?

The animal did go to p2 because it just observed that no screen is present, and it believed that food is present at p2.

Why did it believe that food was present at p2 ?

It believed that food is present at p2 because it already had this belief earlier, and this belief persisted.

When did it start to have this belief ?

It started to have this belief that food is present at p2 when it observed food at p2.

As discussed in Section 3.2.4 in relation to the difference between *rigid* delayed response behaviour (external relationship **EDR1**) and *updating* delayed response behaviour (external relationship **EDR2**), internal (persistency) relationship **IDR2** entails that in case at one point in time the presence of food at p2 is observed, and at a later point in time the absence of food at p2 is observed, then still b1 will persist. For the rigid case, possibly, in addition to the belief that food is present at position p2, another internal property, namely the belief that food is *not* present at position p2 will start to hold after the observation of the absence of food at p2. For the case of updating delayed response behaviour, a slightly more sophisticated option is that, after the absence of food at p2 has been observed, the belief that food is present at p2 will not be forced to hold anymore. This is expressed by the following dynamic relationship, which provides an alternative for **IDR2**:

IDR4

for all time points t1 and t2 with t1<t2
if the internal belief that food is present at position p2 holds at t1,
and between t1 and t2 the animal does not observe that food is not present at position p2,
then the internal belief that food is present at position p2 holds at t2

4 Simulation Experiments

Simulations have been performed in the LEADSTO software environment. An example trace for the external model for stimulus-response behaviour can be seen in Figure 7. Here, time is on the horizontal axis, the state properties are on the vertical axis. A dark box on top of the line indicates that the property is true during that time period, and a light box below the line indicates that the property is false. This trace is based on the external relationship **ESR1**. The values `(0,0,1,1)` have been chosen for the timing parameters `e`, `f`, `g`, and `h`. Initially the animal observes that there is food at position p2 and a screen is present at time point 0. After observing the screen is no longer present the animal performs the action of going to position p2 after time point 2.

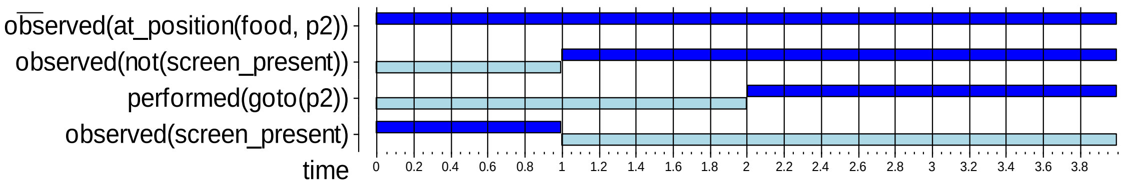

Figure 7 Example trace for the external behavioural view on stimulus-response behaviour

An example trace for the case the internal cognitive model for stimulus-response behaviour can be seen in Figure 8. Initially at time point 0, the animal observes that there is food at position p2 and there is a screen, leading to sensory representations for food at p2, and the presence of the screen. At time point 1 it observes the absence of the screen which leads to animal's internal sensory representation for no screen after time point 2. As a consequence the animal performs the action of going to position p2 at time point 3.

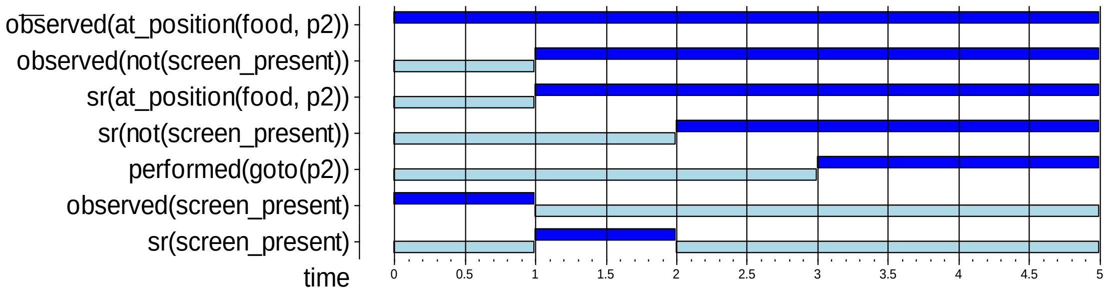

Figure 8 Example trace for the internal cognitive view on stimulus-response behaviour

An example trace for the internal cognitive model for rigid delayed response behaviour is shown in Figure 9. Initially at time point 0, animal observes that food is present at position p2 which leads to the belief that food is present at position p2 at time point 1. Once this belief holds, it persists forever. At time 2, the animal observes that no screen is present, and because at that time point the animal had the belief that food is present at position p2, the animal goes to p2 at time point 2.

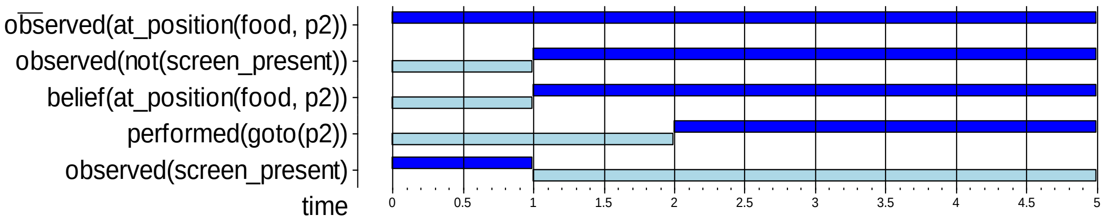

Figure 9 Example trace for the internal cognitive view on rigid delayed response behaviour (scenario 1)

In Figure 10, a different scenario of rigid delayed response behaviour can be seen. In this simulation, it is shown that the animal generates the belief at time point 1 that food is present at position p2 which persists forever, even though at time point 3 the animal observes that *no* food is present at position p2. Because of the rigid behaviour the belief still holds at the time point it is observed that there is no screen anymore. As a consequence of this, the animal goes to position p2 at time point 5, but does not find food there.

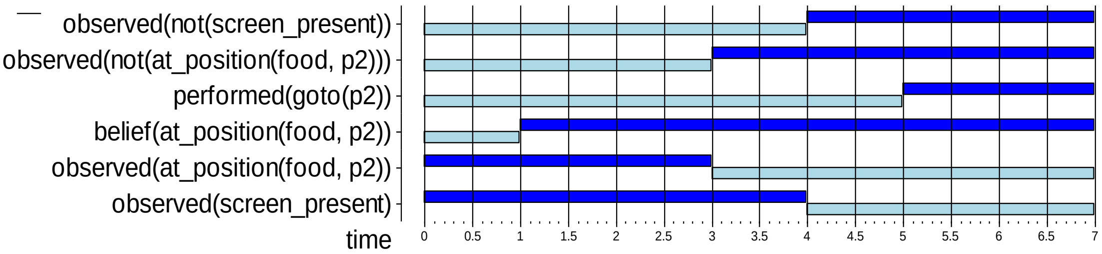

Figure 10 Example for the internal cognitive view on rigid delayed response behaviour (scenario 2)

An example trace for the case of internal cognitive model for updating delayed response behaviour is shown in Figure 11. Initially from time point 0 to 4, the animal observes the presence of food at position p2, leading to the generation of a belief at time point 1, which persists for a longer time. As at the same time it is observed that no screen is present, thus it goes to p2 at time point 2. But because this is an example of updating delayed response behaviour, as soon as animal observes the absence of food at position p2 at time point 4, the belief is withdrawn from time point 5.

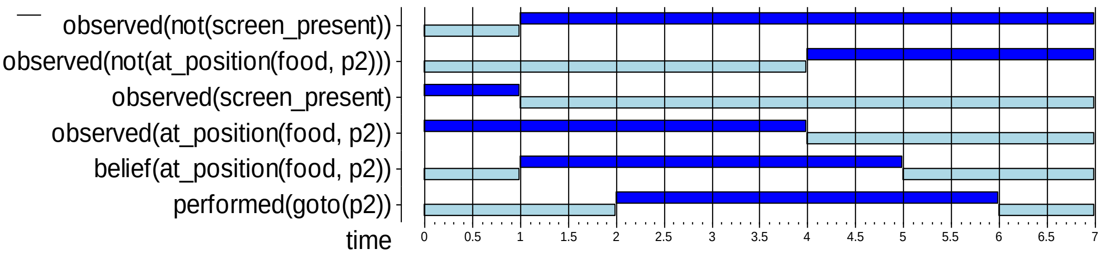

Figure 11 Example for the internal cognitive view on updating delayed response behaviour (scenario 1)

In Figure 12, a different scenario of updating delayed response behaviour is shown. Initially at time point 0 when the animal observes that food is present, its belief starts to hold from time point 1, but after animal observes at time point 4 that *no* food is present at position p2, the belief does not hold anymore from time point 5. When the animal again observes the presence of food at time point 10, the belief starts to hold again and persists for a longer time. As a consequence, when the animal observes the absence of screen at time point 12, it performs the action of going to position p2 at time point 13. At time point 18 when the animal observes that *no* food is present at position p2, the belief of the animal starts to not hold anymore (at time point 19).

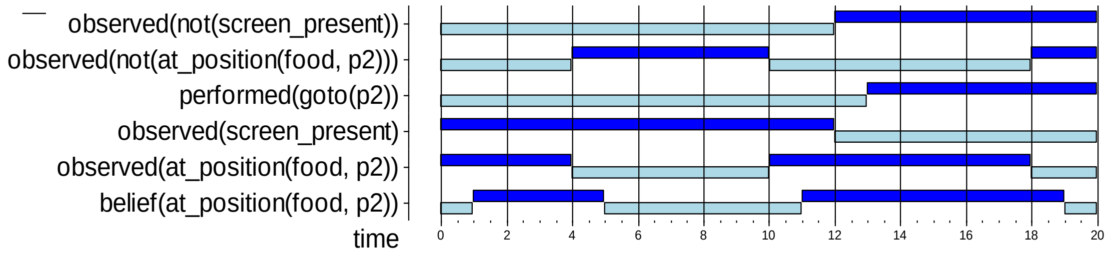

Figure 12 Example for the internal cognitive view on updating delayed response behaviour (scenario 2)

5 Evaluation

In this section, the simulations performed using the model are evaluated with respect to the characteristic patterns identified in Section 1.2. Moreover, the assumptions behind the model are summarised. Finally some possibilities for variations or refinements of the model are discussed.

5.1 Evaluation of the simulations and characteristic patterns

A number of different simulations conducted showed the characteristic patterns identified in Section 1.2. It is easy to perform a few more experiments to check more scenarios.

5.2 Assumptions behind the model

The following assumptions were made.

- Beliefs persist long time periods.
- The animal always wants food (it is given no food during some time before the experiment).
- Precise timings of the steps were neglected.

5.3 Possible variations and refinements

It is not too difficult to refine the model into a more complex model to relax some of the assumptions that were made. For example:

- Beliefs have a decay; after some time they are lost.
- Incorporate motivations such as being hungry.
- Incorporate precise time durations for each of the steps.

6 Further Explorations

A number of further explorations can be made.

More precise timing of decision making and action performance
The model can be refined by incorporating action effects explicitly (for example, self being at position p2), eat actions and more precise timing parameters, both for internal cognitive process steps, and for the actions and their effects. Timing of the internal processes can be taken much faster than the actions.

Defeasible memory: decay of beliefs
Also the phenomenon of defeasibility of memory or decay of beliefs can be explored: after some time duration they are lost. For example after a year you are not able to find back the place where you stored some documents.

Information exchange in a community
Suppose in a community each member maintains beliefs about information about the world (world model) you hear from others, and about topics in which another individual is interested (interest model). If an individual gets information, and it is believed that some other individual is interested in the topic of this information, then the information is communicated to that individual (reactive behaviour). This pattern can be used, for example, to describe the exchange of information in research communities (e.g., Jonker, Lam, and Treur, 2001), or for gossips propagating through communities or social networks.

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